

Name: Gaurika Mittal

Batch: 15

Sap ID: 590034892

Experiment-22

Question

Write a **menu-driven C program** that displays the following menu and uses a **switch statement** to print the selected square/rectangular pattern. Use **nested loops without arrays**. Each pattern should be implemented using a **separate parameterless function with void return type**.

Menu:

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

1. Algorithm

1. Start.
2. Declare the functions `solidSquare()`, `solidRectangle()`, `rowNumber()`, and `columnNumber()` with `void` return type and no parameters.
3. Declare an integer variable `choice`.
4. Display the menu.
5. Accept the user's choice.
6. Use a `switch` statement to perform the selected operation.
7. If choice is 1:
 - o Call `solidSquare()`.
 - o Input the number of rows `n`.
 - o Use nested loops to print `n` rows and `n` columns of `*`.
8. If choice is 2:
 - o Call `solidRectangle()`.
 - o Input rows `m` and columns `n`.
 - o Use nested loops to print `m` rows and `n` columns of `*`.
9. If choice is 3:
 - o Call `rowNumber()`.
 - o Input rows `m` and columns `n`.
 - o Use nested loops.
 - o Print the row number `i` in every column of that row.

10. If choice is 4:
 - Call `columnNumber()`.
 - Input rows `m` and columns `n`.
 - Use nested loops.
 - Print the column number `j` in every position.
11. If choice is 5, exit the program.
12. For any other choice, display "Invalid choice".
13. Repeat the menu until the user selects Exit.
14. Stop.

2. Test-Case Table

Test Case	Choice	Input	Expected Output	Result
1	1	<code>n = 4</code>	4 × 4 solid square	Valid
2	2	<code>m = 3, n = 5</code>	3 × 5 solid rectangle	Valid
3	3	<code>m = 4, n = 4</code>	Rows contain 1s, 2s, 3s, 4s respectively	Valid
4	4	<code>m = 4, n = 4</code>	Each row displays 1 2 3 4	Valid
5	1	<code>n = 1</code>	Single *	Boundary
6	6	-	Invalid choice	Invalid
7	5	-	Program exits	Valid

3. C Program

```
#include <stdio.h>
```

```
void solidSquare();
void solidRectangle();
void rowNumber();
void columnNumber();
```

```
int main()
{
    int choice;

    do
    {
        printf("\n1. Solid Square Pattern");
        printf("\n2. Solid Rectangular Pattern");
        printf("\n3. Row-Number Pattern");
        printf("\n4. Column-Number Pattern");
        printf("\n5. Exit");

        printf("\nEnter your choice: ");
        scanf("%d", &choice);
```

```

switch(choice)
{
    case 1:
        solidSquare();
        break;

    case 2:
        solidRectangle();
        break;

    case 3:
        rowNumber();
        break;

    case 4:
        columnNumber();
        break;

    case 5:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice\n");
}

} while(choice != 5);

return 0;
}

void solidSquare()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void solidRectangle()
{
    int m, n, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);

```

```

for(i = 1; i <= m; i++)
{
    for(j = 1; j <= n; j++)
    {
        printf("* ");
    }
    printf("\n");
}

void rowNumber()
{
    int m, n, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);

    for(i = 1; i <= m; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("%d ", i);
        }
        printf("\n");
    }
}

void columnNumber()
{
    int m, n, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);

    for(i = 1; i <= m; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

```

4. Compilation and Execution

```

(kali㉿kali)-[~]
$ vi hdoc23.c

(kali㉿kali)-[~]
$ gcc hdoc23.c -o hdoc23

(kali㉿kali)-[~]
$ ./hdoc23

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit
Enter your choice: 1
Enter number of rows: 4
* * * *
* * * *
* * * *
* * * *

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit
Enter your choice: 2
Enter number of rows and columns: 3 5

```

```

* * * *
* * * *
* * * *

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit
Enter your choice: 3
Enter number of rows and columns: 4 4
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit
Enter your choice: 4
Enter number of rows and columns: 4 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

```

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

Enter your choice: 1

Enter number of rows: 1

*

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

Enter your choice: 6

Invalid choice

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

Enter your choice: 5

Exiting program...